

University of the Armed Forces “ESPE”

Project’s Name: Prov&Tec Management: Comprehensive Sales and Inventory System

Team Name: ByteWorks

Integrants: Pachacama Andrés

Carlos Parra

Paucar Matias

Problem

Currently, Prov&Tec manages its sales and inventory entirely manually, relying on paper documents (sales slips and order forms) and basic spreadsheets. Lacking technological tools, the company has no precise daily control over its inventory of electrical materials.

The main bottleneck is in the warehouse, where the recording of incoming and outgoing goods is slow and inconsistent. This lack of synchronization and reliable information leads to logistical disarray: staff waste excessive time searching for misplaced products, and—unaware of actual stock levels—the company orders materials it already has on hand, resulting in unnecessary expenses.

Consequently, customer service suffers significantly. Buyers wait 10 to 20 minutes while salespeople physically search for less common items and check paperwork; this causes dissatisfaction and leads many customers to leave the store without completing their purchase. Compounding the issue is an accumulation of slow-moving stock, which generates significant financial losses and limits the owner's management capacity. Therefore, implementing a computer system to centralize and automate these processes is essential.

Overview

The Sales and Inventory Management System for Prov&Tec is an object-oriented software solution designed to centralize and automate the logistical and commercial management of electrical material sales. Its primary purpose is to replace manual methods and paper-based documentation with a digital platform that minimizes errors and provides reliable data for daily operations.

The system will feature secure access with defined user roles (such as salesperson or warehouse manager). Its modules will allow for the recording and updating of customer, product, and supplier information. To streamline customer service, salespeople will be able to quickly process sales, while the system updates inventory levels in real time with every sale or supplier purchase.

Finally, to give the owner greater control over the business, the system will include a transaction history, a reporting module, and an alert system that automatically notifies the user when products reach minimum stock levels. This helps prevent stockouts and facilitates strategic decision-making.

Background

Since its inception, Prov&Tec has operated using a strictly manual management model. Although the company has managed to sustain itself, grow, and overcome significant obstacles—such as financial losses due to stagnant inventory that failed to turn over—its reliance on paper limits its efficiency. Consequently, the company seeks to make a technological leap to modernize its logistics processes, maintain rigorous inventory control, and offer more agile service.

Compounding this situation is the market's digital transition, as traditional processes—such as physical invoicing—rapidly shift toward electronic formats. The need to reduce the excessive administrative paperwork burdening the owner, Ximena Añamisé, presented a clear opportunity for a technical solution.

As Object-Oriented Programming students at the University of the Armed Forces (ESPE) and aspiring software engineers, the development team decided to apply their knowledge to solve this real-world problem. This project allows them to put class and object modeling into practice in a production environment while providing Prov&Tec with a tangible tool to enter the digital age, centralize its data, and streamline its daily operations.